

PRODUCT DESCRIPTION

LucernaPro Epoxy / PU Floor Coating is a two-component solvent-borne protective coating for concrete and cement floors, manufactured so that gloss level, film build, colour and anti-slip additions can be specified per project. The EPOXY grade — a high molecular weight epoxy cured with a polyamide hardener — cures to a dense, hard, high-build film with the highest resistance to compression, abrasion and chemicals, suitable in both atmospheric and immersed service: the standard choice for warehouses, production areas, workshops and garages. The POLYURETHANE (PU) grade is more flexible and markedly more resistant to UV and temperature cycling, and is preferred for loading bays, door thresholds and areas exposed to sunlight or machine vibration. Both are applied in two coats. A self-levelling EPOXY grade at 1.5–3 mm is available to order where a seamless mirror-flat floor is required.

KEY DATA

Type	Two-component (2K) solvent-borne floor coating — epoxy (polyamide-cured) and polyurethane (PU) grades
Mixing ratio	Part A : Part B = 4 : 1 by weight, both grades. Weigh the components; do not estimate by eye. Individual formulations may differ — the product label always governs.
Pot life	Approx. 30–40 minutes at 25°C, shorter when hotter. Mix only what can be applied in that time.
Touch dry	Epoxy grade: 2 hours at 25°C — longer in cool or humid conditions
Overcoating	Epoxy grade: from 5 hours once cured through — typically 4–8 hours depending on temperature, humidity and airflow; overnight recoat is also fine. Apply the second coat cross-wise to the first.
Coats	Two — quoted coverage assumes both coats are applied
Coverage	16 kg set ≈ 50 m ² over two coats (≈ 100 m ² per set per single coat; ≈ 5–7 m ² per kg per coat)
Return to service	Walk-on (careful foot access) 5 hours · light foot traffic 2 days · vehicle traffic, heavy loading and full chemical resistance 7 days
Film properties	Epoxy grade, typical: specular gloss 60° > 80 GU · pencil hardness HB–2H · cross-cut adhesion 100/100 · solids 65 ± 5 % · specific gravity 1.35 ± 0.15
Application	Roller, brush or notched trowel; cut in edges and column bases by brush first
Substrate	Concrete or cement screed, min. 28 days cured, sound, moisture content below 10 %, free of rising moisture. Mechanically prepared by grinding, diamond cup wheel or shot blasting, then vacuumed — sweeping is not enough.
Priming	Prime dusty or highly absorbent substrates with LucernaPro Marine Guard before coating
Finish	Epoxy: gloss. PU: gloss, satin or matt to order.
Colour	Any RAL shade to order — quote the RAL reference with the order
Anti-slip	Graded aggregate can be broadcast into the final coat. Reduces slipperiness; a wet floor is not non-slip.
Packaging	16 kg set — Part A 12.8 kg + Part B 3.2 kg · 4 kg small set to order — Part A 3.2 kg + Part B 0.8 kg
Self-levelling	Epoxy, 1.5–3 mm, notched trowel then de-aerated with a spiked roller. Made to order; application guidance provided to purchasers on request.
Tool cleaning	Thinner, immediately while still wet — cured material cannot be removed. Clean mixing containers before reuse: cured particles from a previous batch cause defects.
Storage	12 months in unopened containers, tightly closed, in a cool, dry, well-ventilated shaded place below 37°C, away from heat and ignition. Keep out of reach of children.

NOTE ON VALUES

Typical values for the standard formulations at 25°C, given as guidance and not as a specification. Drying and cure times shorten in hot weather and lengthen in cool or humid conditions, and coverage depends strongly on substrate porosity. Where the product label differs from this sheet, the label governs.

APPLICATION NOTES

- Surface preparation is the majority of the work. Grind or blast away polished laitance, oil, old paint and all friable material, then vacuum thoroughly. Coatings applied over dust or a weak surface layer delaminate, usually within the first year.
- Moisture limits. Do not apply to wet or green concrete, or over polymer-modified patches, above 5 % moisture; do not apply when air or substrate temperature is within 3°C of the dew point; protect the substrate from condensation from pipes or overhead leaks during application. Moisture trapped under the film is the classic cause of blistering and delamination.
- This is a coating, not a levelling product. The finished film follows the profile of the substrate: waviness, hollows and joints remain visible, and gloss can make them more obvious. Repair before coating — fill cracks and spalls (PatchPro / CreteRevive), level hollows (LevelPro) and allow repairs to cure first.
- Build thickness with mortar, not with resin. For several millimetres of build, trowel a resin-and-graded-quartz epoxy mortar first and finish thin on top — thick pours of neat resin generate exotherm, shrink and crack. A dry, sound, non-chemically-exposed slab can be levelled far more cheaply with a cementitious screed.
- Mix thoroughly to a single uniform colour, scraping the sides and base of the container. Incomplete mixing is the most common cause of patchy, soft or tacky areas.
- Do not thin to extend coverage. Small additions of thinner to adjust viscosity are acceptable; heavier thinning reduces film build and durability proportionally.
- The epoxy grade discolours and dulls under prolonged UV exposure and is intended for interior use. Specify the PU grade for areas exposed to sunlight.

HEALTH AND SAFETY

These products contain organic solvent: they are flammable and give off vapour during application and drying. Apply only with genuinely effective ventilation, use forced-air extraction in enclosed areas, and eliminate all sources of ignition, including smoking, from the work area and adjacent spaces — never work in a sealed room. Wear rubber gloves, safety glasses and an organic-vapour respirator. The hardener may cause skin sensitisation: avoid skin contact and wash off immediately with soap and water, never with thinner. Refer to the Safety Data Sheet for the grade supplied before use.

Given in good faith on the basis of our current knowledge and applying to the product as supplied. Site conditions, substrate condition and workmanship are outside our control and no warranty of result is given or implied; users should satisfy themselves that the product suits the intended use, by trial on site if necessary.